Zomato Delivery Operations

Analysis & Predictive Modeling Project Report

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Tools: Python (pandas, scikit-learn, seaborn) | Google Colab

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1. Executive Summary

This report analyzes 45,584 Zomato food-delivery orders to evaluate delivery performance, optimize routing, understand the drivers of customer wait times, surface operational patterns, and build a predictive model for delivery time. The dataset spans 11 Feb 2022 to 6 Apr 2022 and covers 1,320 unique delivery partners across Metropolitan, Urban, and Semi-Urban city types.

Key headline numbers:

Average delivery time: 26.3 minutes (median 26 minutes)
Average delivery-partner rating: 4.63 / 5
Average trip distance: 9.7 km
Best predictive model: Random Forest Regressor — $R^2 = 0.84$, MAE = 3.0 minutes
The analysis shows that road-traffic density, weather conditions, festival days, and the number of bundled deliveries per trip are the strongest drivers of delivery delay — more influential than raw distance alone. Section 8 lists concrete operational recommendations based on these findings.

2. Introduction & Objectives

Zomato's delivery dataset provides a comprehensive view of delivery operations, including delivery-person details, order timestamps, weather conditions, traffic density, and location coordinates. The objective of this project is to explore this data to optimize delivery routes and enhance customer satisfaction. The analysis was carried out against five objectives:

- Delivery Performance Analysis — evaluate delivery-person ratings, delivery times, and vehicle condition.
- Route Optimization — use location and traffic data to identify where delivery times can be reduced.
- Customer Experience Enhancement — quantify how weather and festival seasons affect delivery time.
- Operational Insights — examine order volumes, order types, and multiple-delivery trips.
- Predictive Analytics — build a model to forecast delivery time from order-level factors.

3. Dataset Overview

The raw dataset contains 45,584 rows and 20 columns. Key fields include:

Field	Description
Delivery_person_Ratings, Age	Delivery partner's rating and age
Restaurant / Delivery lat-long	Used to derive trip distance
Order_Date, Time_Orderd, Time_Order_picked	Used to derive order hour and food-prep time
Weather_conditions, Road_traffic_density	Environmental and traffic context
Vehicle_condition, Type_of_vehicle	Condition (0–3) and vehicle type used

Field	Description
Type_of_order, multiple_deliveries, Festival, City	Order and operational context
Time_taken (min)	Target variable — total delivery time

3.1 Data Cleaning

- Missing values were present in Age (1,854), Ratings (1,908), Time_Orderd (1,731), Weather_conditions (616), Road_traffic_density (601), multiple_deliveries (993), Festival (228), and City (1,200) — handled via median/mode imputation or row exclusion depending on the analysis.
- Trip distance was derived from restaurant and delivery-location coordinates using the Haversine formula; a small number of implausible values (>50 km or near-zero) were treated as missing.
- Food preparation time was derived as the gap between Time_Orderd and Time_Order_picked, adjusting for orders that cross midnight.
- Order hour and day-of-week were extracted from the timestamp fields to support demand-pattern analysis.

4. Delivery Performance Analysis

Delivery-person rating shows a moderate negative correlation with delivery time (r = –0.34): higher-rated partners tend to complete deliveries faster. Vehicle condition also matters — orders assigned to the worst vehicle-condition category (0) take noticeably longer on average than better-maintained vehicles.

Vehicle Condition	Avg. Delivery Time (min)
0 (Poor)	30.1
1	24.4
2	24.5
3 (Best)	26.5

Across the 1,320 delivery partners in the dataset, those with 5+ completed orders and consistently high ratings also tend to post the lowest average delivery times, confirming ratings are a useful proxy for operational performance and a reasonable input for incentive or training programs.

5. Route Optimization

Trip distance is positively but only moderately correlated with delivery time (r = 0.32, average distance 9.7 km), which indicates that distance alone is not the dominant driver of delay — road-traffic density has a much larger effect at a similar distance.

Traffic Density	Avg. Delivery Time (min)
Low	21.3

Traffic Density	Avg. Delivery Time (min)
Medium	26.7
High	27.2
Jam	31.2

Delivery time rises by roughly 10 minutes from Low to Jam traffic conditions. This confirms that traffic-aware routing and dispatch — rather than pure shortest-distance routing — would meaningfully cut delivery times, especially during identified peak-traffic windows.

City Type	Avg. Delivery Time (min)
Urban	23.0
Metropolitan	27.3
Semi-Urban	49.7

Semi-Urban deliveries take almost twice as long on average as Urban ones, flagging Semi-Urban zones as the top priority for additional delivery partners, better route planning, or revised delivery-time expectations shown to customers.

6. Customer Experience Enhancement

Weather has a clear effect on delivery time: Sunny conditions see the fastest average deliveries, while Fog and Cloudy conditions are slowest.

Weather	Avg. Delivery Time (min)
Sunny	21.9
Stormy	25.9
Sandstorms	25.9
Windy	26.1
Fog	28.9
Cloudy	28.9

Festival days show the largest single effect of any factor analyzed: average delivery time jumps from 26.0 minutes on non-festival days to 45.5 minutes on festival days — a 75% increase, driven by demand spikes and congestion.

Recommendation: during adverse weather or festival periods, proactively widen the ETA shown to customers and trigger order-status notifications rather than holding to a standard promised time, which is the more common cause of customer dissatisfaction than the delay itself.

7. Operational Insights

Order volume is fairly evenly spread across the four order types (Buffet, Drinks, Meal, Snack), with average delivery time nearly identical across types (26.2–26.4 minutes) — order type itself is not a meaningful driver of delay.

The number of bundled deliveries per trip has a large effect: single-order trips average 22.9 minutes, while trips carrying 3 simultaneous deliveries average 47.8 minutes — more than double.

Deliveries per Trip	Avg. Delivery Time (min)
1 order	22.9
2 orders	26.9
3 orders	40.5
4 orders	47.8

Order volume peaks in the evening, with the busiest hours between 5 PM and 10 PM (each hour handling 4,000+ orders in the dataset), pointing to when rider staffing should be at its highest.

Recommendation: cap bundled deliveries at 2 orders per trip during evening peak hours unless the added orders are on the same route, since the time cost of a 3rd or 4th bundled order outweighs the throughput gain in most cases.

8. Predictive Analytics: Delivery Time Forecasting

Three regression models were trained to forecast Time_taken (min) using distance, delivery-partner rating and age, vehicle condition, multiple-deliveries count, order hour, prep time, weather, traffic density, order type, vehicle type, festival flag, and city as inputs (41,213 complete rows after cleaning; 80/20 train-test split).

Model	MAE (min)	RMSE (min)	R^2
Linear Regression	4.75	5.98	0.585
Gradient Boosting	3.49	4.36	0.780
Random Forest	3.01	3.75	0.837

The Random Forest model performed best, explaining roughly 84% of the variance in delivery time with an average error of about 3 minutes — accurate enough to support real-time ETA estimation. Feature-importance analysis (see notebook) confirms that trip distance, road-traffic density, weather condition, and multiple-deliveries count are the leading predictors, consistent with the descriptive findings above.

This model can be deployed as a lightweight ETA service: given an order's distance, current weather/traffic conditions, and rider profile, it returns a delivery-time estimate before the order is dispatched, enabling more accurate customer-facing ETAs and proactive staffing decisions.

9. Recommendations

- Prioritize vehicle maintenance/replacement for the poor-condition (0) fleet segment — it shows the highest average delivery time.
- Adopt traffic-aware dispatch and routing rather than pure shortest-distance routing, especially during High/Jam traffic windows.
- Increase delivery-partner capacity in Semi-Urban zones, where average delivery time is nearly double that of Urban zones.
- Widen customer-facing ETAs and send proactive notifications during Fog/Cloudy weather and festival periods.
- Limit bundled deliveries to 2 per trip during evening peak hours (5–10 PM) unless stops are on the same route.
- Deploy the Random Forest model as a real-time ETA estimator to improve the accuracy of promised delivery times.

10. Conclusion

This analysis confirms that delivery time in the Zomato dataset is driven less by raw distance and more by operational and environmental context — traffic density, weather, festival timing, and the number of bundled deliveries per trip. A Random Forest model built on these factors forecasts delivery time within about 3 minutes on average ($R^2 = 0.84$), providing a practical foundation for real-time ETA estimation and for prioritizing operational improvements such as fleet maintenance, Semi-Urban capacity, and peak-hour dispatch rules.